(c) (i) How does robot-oriented programming differ from object-oriented programming? Discuss. 5

(ii) What is object recognition in image processing? Explain it and enumerate its importance. 5



Printed Pages – 8

# 548311 (37)

## ME/M Tech (3rd Semester)

## Examination, Nov-Dec., 2014

## Branch : Mechanical Engineering

## Specialization : Design

## ROBOTICS

*Time Allowed : Three Hours*

*Maximum Marks : 100*

*Minimum Pass Marks : 40*

**Note :** Answer all the questions. Each question carries equal marks. Give suitable diagrams and flow-charts or block diagrams wherever needed. Answer any two parts from each question

**Q. 1.** (a) How many degrees of freedom are required in a robot manipulator? Also explain spatial, planar, and redundant manipulators on the basis of degree of freedom. 10

(b) Describe the basic arm configurations of a robotic manipulator. What are the differences between polar-arm and articulated-arm type configurations? Discuss them and draw the suitable views of the work spaces for these two arm configurations. **10**

(c) A cylindrical robot has a prismatic joint with a range of travel of 1500 mm. The control memory for this joint has 12 bit capacity. The associated mechanical in accuracies with the arm about the mean position gives a standard deviation of 0.01 mm, which is equal in all the directions. Determine :

(i) The control resolution for the axis.

(ii) The spatial resolution for the given joint.

(iii) The accuracy defined, and

(iv) The repeatability of the robot link. **10**





**Q. 2.** (a) Using DH notation algorithm, is it possible to have a link with zero link-length whereas the physical link on the manipulator will have a finite link-length? Justify your answer. Also discuss the importance of choosing a frame assignment for an n-DOF manipulator which gives a maximum number of zero-link parameters. **10**

(b) Frame {1} is to be rotated with respect to frame {0} about x-axis by an angle of $\pi/3$. The position of the origin of frame {1} as seen from frame {0} is given by ${}^0D_1 = [7 \quad 5 \quad 7]^T$. Determine the transformation matrix ${}^0T_1$ to describe the frame {1} relative to frame {0}. Further, determine the description of a point Q is frame {0} if ${}^1Q =$

$[2\ 4\ 6]^T$, and using the matrix ${}^0T_1$, determine the description of frame {0} relative to frame {1}. **10**

(c) Describe the various factors/parameters on which the number of solutions to a given inverse kinematics model depends. Why are the solutions to inverse kinematics problems generally difficult? Why closed form solutions are preferred over other form of solutions to inverse kinematic problems? **10**

**Q. 3.** (a) Obtain the Lagrange Function for a planar, two degree of freedom manipulator with both rotary joints and joint variables $\theta_1$ and $\theta_2$. The link lengths are $\ell_1$ and $\ell_2$ with masses $m_1$ and $m_2$ respectively. Also state clearly the assumptions taken in above, if any. **10**





(b) An n-degree of freedom robotic manipulator is to be required to perform at very low velocities. What simplification will result from this condition in the LE formulation? Further, if a manipulator is to be used in a gravity-free condition, what will be the effect of this situation on the dynamic model? Also discuss the forces applicable and significant under these conditions. **10**

(c) Discuss the following :

(i) LE formulation characteristics behaviour **5**

(ii) Comparison among NE, LE and GD formulations **5**

**Q. 4.** (a) Discuss the importance of actuators in the robot design. Also discuss the features of DC, AC and stepper motor drives. **10**

(b) (i) What are the general considerations in trajectory planning of a manipulator? **5**

(ii) The revolute joint of an articulated point to point robot traverses from an initial position of $10^0$ to $30^0$ final position within 2 seconds. Assume a third-degree polynomial and start-off acceleration 4 degree/sec, determine the deceleration at the end of 2 seconds. **5**

(c) The control function of a certain robot manipulator is described by the differential equation specifying the position of the output link as a function of time,

$$3.2\frac{d^2y}{dt^2} + 29.4\frac{dy}{dt} + 37.6 = X(t)$$

where, y gives the output response and X is the function describing the force applied. Determine the following :





(i) The characteristic equation and its roots **3**

(ii) The transfer function **2**

(iii) The natural frequency and resonant frequency **3**

(iv) The damping ratio. **2**

**Q. 5.** (a) Discuss all about the robotic vision system, its architecture, components and working principle briefly giving suitable diagram. **10**

(b) (i) Why does sensor selection important in robot design? What are the factors or parameters to be considered for sensor selection? **5**

(ii) Discuss briefly any five non-industrial applications of robot. **5**